

Full Publication List

First-Author Journal Articles:

1. **Liu H**, Grouza V, Tuznik M, Bagheri H, Peterson A, Rudko D. (2022) Self-Labelled Encoder-Decoder (SLED) for Multi-Echo Gradient Echo-Based Myelin Water imaging. *NeuroImage*. 264, p119717.

<https://doi.org/10.1016/j.neuroimage.2022.119717>.

2. **Liu H***, Xiang QS, Tam R, Kozlowski P, Li DKB, Mackay AL, Kramer, JK and Laule C. (2021) A data-driven T_2 relaxation analysis approach for myelin water imaging: Spectrum analysis for multiple exponentials via experimental condition oriented simulation (SAME-ECOS). *Magn Reson Med*. 87(2), pp.915-931.

<https://doi.org/10.1002/mrm.29000>

3. **Liu H***, Xiang, QS, Tam R, Dvorak AV, MacKay AL, Kolind SH, Traboulsee A, Vavasour IM, Li DKB, Kramer JK, and Laule C. (2020) Myelin water imaging data analysis in less than one minute. *NeuroImage*, 210, p.116551.

<https://doi.org/10.1016/j.neuroimage.2020.116551>

4. **Liu H**, Ljungberg E, Dvorak AV, Lee LE, Yik JT, MacMillan EL, Barlow L, Li DKB, Traboulsee A, Kolind SH, Kramer JKL, and Laule C. (2020) Myelin water fraction and intra/extracellular water geometric mean T_2 normative atlases for the cervical spinal cord from 3T MRI. *Journal of Neuroimaging*, 30(1), pp.50-57.

<https://doi.org/10.1111/jon.12659>

5. **Liu H**, Rubino, C., Dvorak AV, Jarrett M, Ljungberg E, Vavasour IM, Lee LE, Kolind SH, MacMillan EL, Traboulsee A, Lang DJ, Rauscher A, Li DKB, MacKay AL, Boyd LA, Kramer JKL, Laule C. (2019) Myelin water atlas: a template for myelin distribution in the brain. *Journal of Neuroimaging*, 29(6), pp.699-706.

<https://doi.org/10.1111/jon.12657>

6. **Liu H***, MacMillan EL, Jutzeler CR, Ljungberg E, MacKay AL, Kolind SH, Madler B, Li DKB, Dvorak MF, Curt A, Laule C, and Kramer JL. (2017) Assessing structure and function of myelin in cervical spondylotic myelopathy: evidence of demyelination. *Neurology*, 89(6), pp.602-610.

<https://doi.org/10.1212/WNL.0000000000004197>

Co-Author Journal Articles:

1. Grouza V, Bagheri H, **Liu H**, Tuznik M, Wu Z, Robinson N, Siminovitch KA, Peterson AC, Rudko DA. Ultra-high-resolution mapping of myelin and g-ratio in a panel of Mbp enhancer-edited mouse strains using microstructural MRI. *NeuroImage*. 2024 Oct 15;300:120850.

<https://doi.org/10.1016/j.neuroimage.2024.120850>

2. Michell-Robinson MA, Watt KE, Grouza V, Macintosh J, Pinard M, Tuznik M, Chen X, Darbelli L, Wu CL, Perrier S, Chitsaz D, Uccelli N, **Liu H**, Coulombe G, Rudko D, Bernard G. Hypomyelination, hypodontia and craniofacial abnormalities in a Polr3b mouse model of leukodystrophy. *Brain*. 2023 Aug 28, awad249.

<https://doi.org/10.1093/brain/awad249>

3. Faw TD, Lakhani B, Schmalbrock P, Knopp MV, Lohse KR, Kramer JL, **Liu H**, Nguyen HT, Phillips EG, Bratasz A, Fisher LC. (2021) Eccentric rehabilitation induces white matter plasticity and sensorimotor recovery in chronic spinal cord injury. *Experimental neurology*. 346, p.113853.

<https://doi.org/10.1016/j.expneurol.2021.113853>

4. Lee LE, Vavasour IM, Dvorak AV, **Liu H**, Abel S, Johnson P, Ristow S, Au S, Laule C, Tam R, Li DKB, Cross H, Ackermans N, Schabas AJ, Chan J, Sayao A, Devonshire V, Carruthers R, Traboulsee A, and Kolind, SH. (2021) Cervical cord myelin abnormality is associated with clinical disability in multiple sclerosis. *Multiple Sclerosis Journal*. 27(14), pp. 2191–2198.

<https://doi.org/10.1177/13524585211001780>

5. Dvorak AV, Swift-LaPointe T, Vavasour IM, Lee LE, Abel S, Russell-Schulz B, Graf C, Wurl A, **Liu H**, Laule C, Li DKB, Traboulsee A, Tam R, Boyd LA, Mackay AL, and Kolind SH. (2021) An atlas for human brain myelin content throughout the adult life span. *Scientific reports*, 11(1), pp.1-13.

<https://doi.org/10.1038/s41598-020-79540-3>

6. Morris SR, Holmes RD, Dvorak AV, **Liu H**, Yoo Y, Vavasour IM, Mazabel S, Mädler B, Kolind SH, Li DKB and Siegel L, Beaulieu C, MacKay A, and Laule C. (2020) Brain myelin water fraction and diffusion tensor imaging atlases for 9-10 year-old children. *Journal of Neuroimaging*, 30(2), pp.150-160.

<https://doi.org/10.1111/jon.12689>

7. Dvorak AV, Ljungberg E, Vavasour IM, **Liu H**, Johnson P, Rauscher A, Kramer JL, Tam R, Li DKB, Barlow L, Laule C, MacKay A, Traboulsee A, Kozlowski, Kolind SH. (2019) Rapid myelin water imaging for the assessment of cervical spinal cord myelin damage. *NeuroImage: Clinical*, 23, p.101896. <https://doi.org/10.1016/j.nicl.2019.101896>

First-Author Conference Proceedings:

1. **H. Liu**, N. Wiley, S. Balaji, M. Poorman, F. Padormo, R. Azeredo Gomes Teixeira, A. Dvorak, S. Williams, S. C. Deoni, A. Traboulsee, S. H. Kolind; (2024) Detecting myelin loss at 0.064T using a novel myelin water imaging technique: A Multiple Sclerosis Case Study. *Proceedings of Radiological Society of North America (RSNA)*.

2. **Liu H**, Vladimir Grouza, Marius Tuznik, David Rudko. (2022) Application of SAME-ECOS to 7T gradient-echo based myelin water imaging: a comparison of model-free and model-based approaches. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 30:4970.

<https://index.mirasmart.com/ISMRM2022/PDFfiles/4970.html>

3. **Liu H**, Xiang QS, Tam R, Kozlowski P, Li D, MacKay A, Kramer J, Laule C. (2020) A novel T2 relaxation analysis method SAME-ECOS: Spectrum Analysis for Multiple Exponentials via Experimental Condition Oriented Simulation. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 28:3157.

<https://archive.ismrm.org/2020/3157.html>

4. **Liu H**, Xiang QS, Tam R, MacKay A, Kramer JK, Laule C. (2019) 150× acceleration of myelin water imaging data analysis by a neural network. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 27:0152.

<https://archive.ismrm.org/2019/0152.html>

5. **Liu H**, Tam R, Kramer JK, Laule C. (2019) Analyzing multi-exponential T2 decay data using a neural network. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 27:4886.

<https://archive.ismrm.org/2019/4886.html>

6. **Liu H**, Tam R, Kramer JK, Laule C. (2019) Myelin Water Imaging Data Post-processing: A Deep Learning Approach. *International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on Machine Learning Part II*.

<https://cds.ismrm.org/protected/Machine18II/program/abstracts/posters/LiuH.pdf>

7. **Liu H**, Rubino C, Jarrett M, Ljungberg E, Vavasour I, Kolind SH, MacMillan EM, Traboulsee T, Lang D, Rauscher A, Li D, MacKay A, Boyd L, Kramer JK, Laule C. (2019) Myelin water atlas: a template for myelin distribution in the brain. *Americas Committee for Treatment and Research in Multiple Sclerosis (ACTRIMS)*. Volume 25 Issue 1_suppl, P187.

<https://doi.org/10.1177/1352458519843086> (@ P187)

8. **Liu H**, Rubino C, Jarrett M, Ljungberg E, Vavasour I, Kolind SH, MacMillan EM, Traboulsee T, Lang D, Rauscher A, Li D, MacKay A, Boyd L, Kramer JK, Laule C. Myelin water atlas: a template for myelin distribution in the brain. (2018) *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 26:0264.

<https://archive.ismrm.org/2018/0264.html>

9. **Liu H**, Ljungberg E, MacMillan EM, Barlow L, Kolind SH, Kramer JK, Laule C. (2017) Myelin water atlas for cervical spinal cord: A template for spinal cord pathway myelin microstructure. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 25:5062.

<https://archive.ismrm.org/2017/5062.html>

10. **Liu H**, Ljungberg E, MacMillan EM, Barlow L, Kolind SH, Kramer JK, Laule C. (2017) Quantitative measurement of myelin in distinct pathways of the spinal cord: Implications for assessing neuropathic pain after spinal cord injury. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 25:0913.

<https://archive.ismrm.org/2017/0913.html>

11. **Liu H**, Ljungberg E, MacMillan EM, Barlow L, Kolind SH, Kramer JK, Laule C. (2017) Assessing myelin in white matter pathways in the spinal cord by myelin water imaging: Implications for investigating neuropathic pain. *Proceedings of International Society for Magnetic Resonance in Medicine Workshop on Quantitative MRI in White Matter Disorders: Useful, Usable, Used?*

https://cds.ismrm.org/protected/WhiteMatter17/Program/abstracts/liu_assessing_myelin_in_white_matter.pdf

12. **Liu H**, Ljungberg E, MacMillan EM, Barlow L, Kolind SH, Kramer JK, Laule C. (2017) Myelin water atlas for white matter pathways in the cervical spinal cord. *Proceedings of International Society for Magnetic Resonance in Medicine Workshop on Quantitative MRI in White Matter Disorders: Useful, Usable, Used?* , https://cds.ismrm.org/protected/WhiteMatter17/Program/abstracts/Liu_Myelin_water_atlas.pdf

13. **Liu H**, MacMillan EL, Ljungberg E, MacKay AL, Kolind SH, Mädler B, Li DKB, Dvorak MF, Curt A, Laule C, Kramer JLK. (2016) Assessing Structure and Function of Myelin in Cervical Spondylotic Myelopathy: Evidence of Focal Demyelination in the Dorsal Column. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 24:0849.

<https://archive.ismrm.org/2016/0849.html>

14. **Liu H**, Ljungberg E, MacMillan EM, Barlow L, Kolind SH, Laule C, Kramer JK. (2016) Objective measurement of myelin in the dorsal column and spinothalamic tract: Implications for assessing neuropathic pain after spinal cord injury. *Proceedings of 37th Annual Scientific Meeting of the Canadian Pain Society*.

Corresponding-Author Conference Proceedings:

1. Yik J, Tam R, Kramer JK, Laule C, **Liu H***. (2019) Spinal cord grey matter segmentation using an Encoder-Decoder Convolutional Neural Network. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 27:4795.

<https://archive.ismrm.org/2019/4795.html>

2. Yik J, Tam R, Rubino C, Boyd L, Li DKB, Laule C, **Liu H***. (2019) Feasibility of brain white matter segmentation on multi-echo T2-weighted images without registration: a neural network approach. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 27:4796.

<https://archive.ismrm.org/2019/4796.html>

Co-Author Conference Proceedings

1. Grouza V, Bagheri H, Tuznik M, **Liu H**, Peterson A, Rudko D. (2022) Sensitive Quantification of Hypomyelination and Axon g-ratio Using Ultra High Resolution 7T Multi-Echo Gradient Echo MRI with BSS-rPCA. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 30:0524.

<https://index.mirasmart.com/ISMRM2022/PDFfiles/0524.html>

2. Balaji S, Dvorak A, Vavasour IM, Poorman ME, **Liu H**, Ljungberg E, Williams S, Deoni S, Laule C, Li DKB, Moore GRW, Mackay A, Kolind SH. (2022) Postmortem Multiple Sclerosis Brain Lesion Identification at Low Field. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 30:1518.

<https://index.mirasmart.com/ISMRM2022/PDFfiles/1815.html>

3. Joseph TS, **Liu H**, Kolind SH, Zhao G, Carruthers R, Schabas A, Sayao A, Devonshire V, Tam R, Moore GRW, Li DKB, Traboulsee A, Vavasour IM, Laule C. (2022) Multiple sclerosis T2 lesion volume correlates weakly with myelin water fraction.

<https://index.mirasmart.com/ISMRM2022/PDFfiles/0202.html>

4. Joseph TS, **Liu H**, Kolind SH, Zhao G, Sun P, Carruthers R, Schabas A, Sayao A, Devonshire V, Tam R, Moore GRW, Li DKB, Traboulsee A, Vavasour IM, Laule C. (2022) Myelin water and multi-shell diffusion imaging provide unique information about multiple sclerosis myelin and axonal damage. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 30:0204.

<https://index.mirasmart.com/ISMRM2022/PDFfiles/0204.html>

5. Joseph TS, **Liu H**, Kolind SH, Zhao G, Sun P, Carruthers R, Schabas A, Sayao A, Devonshire V, Tam R, Moore GRW, Li DKB, Traboulsee A, Vavasour IM, Laule C. (2022) Correlations of inflammation metrics from multi echo T2 relaxation and multi-shell diffusion imaging in multiple sclerosis. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 30:0883.

<https://index.mirasmart.com/ISMRM2022/PDFfiles/0883.html>

6. Joseph TS, **Liu H**, Dvorak AV, Zhao G, Traboulsee A, Kolind SH, Carruthers R, Schabas A, Sayao A-L, Devonshire V, Tam R, Moore GRW, Li DKB, Vavasour IM, Laule C. (2021) A new machine learning approach for myelin water imaging analysis in multiple sclerosis. *Proceedings of 37th Congress of the European Committee for Treatment and Research in Multiple Sclerosis (ECTRIMS)*.

7. Russell-Schulz B, Vavasour IM, **Liu H**, Laule C, Sparrey C, Mackay A, Sawatzky B. (2021) Myelin water imaging of Amyoplasia: Comparison to a control atlas. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 28:1435.

<https://archive.ismrm.org/2020/1435.html>

8. Swift-LaPointe T, Vavasour IM, Lee LE, Abel S, Russell-Schulz B, Graf C, Wurl A, **Liu H**, Laule C, Li DKB, Traboulsee A, Tam R, Boyd L, Mackay AL, Kolind S, Dvorak AV. (2020). Uncharted Waters: Insights From A 100-Subject Myelin Water Imaging Atlas. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 28:1441.

<https://archive.ismrm.org/2020/1441.html>

9. Vavasour IM, Lee LE, Dvorak AV, **Liu H**, Abel S, Johnson P, Ristow S, Laule C, Tam R, Li DKB, Ackermans N, Schabas AJ, Chan J, Sayao A-L, Devonshire V, Carruthers R, Traboulsee A, Kolind SH. (2020) Cervical cord myelin abnormality is better correlated than cord area with clinical disability in multiple sclerosis. Poster Presentation. *Americas Committee for Treatment and Research in Multiple Sclerosis (ACTRIMS)*.

<https://doi.org/10.1177/1352458520917096> (@ P153)

10. Morris S, Holmes RD, Dvorak A, **Liu H**, Vavasour IM, Mazabel S, Mädler B, Kolind S, Li DKB, Siegel L, Beaulieu C, MacKay A, Laule C. (2019). A pediatric brain template for myelin water fraction and diffusion tensor imaging. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 27:0118.

<https://archive.ismrm.org/2019/0118.html>

11. Dvorak A, Johnson P, **Liu H**, Ljungberg E, Vavasour IM, Kramer JK, MacKay A, Laune C, Cashman N, Bremberg H, Kolind SH. (2019). Quantitative MRI with spatiotemporal normalization can detect longitudinal changes in primary lateral sclerosis cervical spinal cord. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 27:0529.

<https://archive.ismrm.org/2019/0529.html>

12. Dvorak A, **Liu H**, Ljungberg E, Vavasour IM, Lee LE, Mackay AL, Tam R, Li DKB, Rauscher A, Laule C, and Kolind SH. (2019) Multivariate template creation of a myelin water brain atlas with GRASE and mcDESPOT. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 28:1435.

<https://archive.ismrm.org/2019/0424.html>

13. Lee LE, Dvorak AV, **Liu H**, Abel S, Johnson P, Vavasour IM, Laule C, Tam R, Li DKB, Chan J, Carruthers R, Traboulsee A, Kolind SH. Does myelin content heterogeneity in the spinal cord reflect disability in multiple sclerosis? *Proceedings of the International Society of Magnetic Resonance in Medicine*. 27:0524.

<https://archive.ismrm.org/2019/0524.html>

14. Morris S, Holmes RD, Dvorak A, **Liu H**, Vavasour IM, Mazabel S, Mädler B, Kolind S, Li DKB, Siegel L, Beaulieu C, MacKay A, Laule C. (2019) A Pediatric Atlas for Multiple Sclerosis Myelin Water and Diffusion Tensor Imaging Studies. *Americas Committee for Treatment and Research in Multiple Sclerosis (ACTRIMS)*. Volume 25 Issue 1_suppl, P156.

<https://doi.org/10.1177/1352458519843086> (@ P156)

15. Lee LE, Vavasour IM, Dvorak AV, **Liu H**, Abel S, Johnson P, Ristow S, Au S, Laule C, Tam R, Li DKB, Cross H, Ackermans N, Schabas AJ, Chan J, Sayao A, Devonshire V, Carruthers R, Traboulsee A, and Kolind, SH. (2019) Associations between Myelin Damage in the Cervical Spinal Cord and Disability in Relapsing and Progressive Multiple Sclerosis. *Americas Committee for Treatment and Research in Multiple Sclerosis (ACTRIMS)*. Volume 25 Issue 1_suppl, P159.

<https://doi.org/10.1177/1352458519843086> (@ P159)

16. Dvorak AV, Ljungberg E, Vavasour IM, **Liu H**, Johnson P, Rauscher A, Kramer JL, Tam R, Laule C, Li DKB, MacKay A, Traboulsee A, Kozlowski, Kolind SH. (2019) Pilot Assessment of Cervical Spinal Cord Myelin Content and Tissue Integrity in Neuromyelitis Optica Spectrum Disorder. *Americas Committee for Treatment and Research in Multiple Sclerosis (ACTRIMS)*. Volume 25 Issue 1_suppl, P192.

<https://doi.org/10.1177/1352458519843086> (@ P192)

17. Laule C, Vavasour IM, Kolind SH, Bjarnason TA, Zhang J, Lang DJM, **Liu H**, Ljungberg E, Tam T, MacMillan EL, Kramer JKL, Sirrs S, Kozlowski P, Rauscher A, Boyd L, Moore GRW, Traboulsee AL, Li DKB, and MacKay A. (2018) Quantitative MRI with spatiotemporal normalization can detect longitudinal changes in primary lateral sclerosis cervical spinal cord. *Proceedings of International Society for Magnetic Resonance in Medicine (ISMRM)*. 25:4196.

<https://archive.ismrm.org/2017/4196.html>

18. Laule C, Vavasour IM, Kolind SH, Zhang J, Lang DJM, **Liu H**, Ljungberg E, Tam R, MacMillan EL, Kramer JK, Sirrs SM, Traboulsee AL, Boyd L, Kozlowski P, Rauscher P, Moore GRW, Li DKB, MacKay AL. (2018) Myelin Water Imaging: Assessment of Myelination with MRI. *Proceedings of 56th Annual Meeting of the American Society of Neuroradiology (ASNR)*.

19. Laule C, Morris SM, Prevost V, **Liu H**, Kozlowski P, Yung A, Bauman A, Samadi F, Fournier C, Alundino A, Parker L, Dong K, Streijger F, Moore GRW, Kwon B. (2018) Quantitative 7T magnetic resonance imaging and histologic analysis of post-mortem human spinal cord injury specimens. *Proceedings of Society for Neuroscience*.

20. Dvorak A, Ljungberg E, Vavasour IM, **Liu H**, Rauscher A, Kramer J, Laule C, MacKay AL, Traboulsee AL, Kozlowski P, Kolind SH. (2017) GRASE myelin water imaging in the cervical spinal cord of neurodegenerative disease patient populations. *Proceedings of 34th Annual Scientific Meeting of the European Society for Magnetic Resonance in Medicine and Biology (EBMRMB)*.

http://esmrmb.org/app/uploads/ESMRMB_2017_Meeting_Guide_WEB_v2.pdf (@ 442)

21. Dvorak A, Ljungberg E, Vavasour IM, **Liu H**, Rauscher A, Kramer J, Laule C, MacKay AL, Traboulsee AL, Kozlowski P, Kolind SH. (2017) Accelerated myelin water imaging for assessment of cervical spinal cord demyelination in multiple sclerosis and neuromyelitis optica spectrum disorder. *Congress of the European Committee for Treatment and Research in Multiple Sclerosis (ECTRIMS)*.